DELHI PUBLIC SCHOOL JANKIPURAM, LUCKNOW

SESSION 2025-2026: PRE BOARD -I

SUBJECT: ARTIFICIAL INTELLIGENCE(417)

SET - I

TIME: 2 Hrs. **M.M.:50**

Date: _____ **Name:** _____ **Class/ SecX**_____

General Instructions:

1. Please read the instructions carefully.
2. This Question Paper consists of **21 questions** in two sections: Section A & Section B.
3. Section A has Objective type questions whereas Section B contains Subjective type questions.
4. **Out of the given (5 + 16 =) 21 questions, a candidate has to answer (5 + 10 =) 15 questions in the allotted (maximum) time of 2 hours.**
5. All questions of a particular section must be attempted in the correct order.
6. **SECTION A - OBJECTIVE TYPE QUESTIONS (24 MARKS):**

i. This section has 05 questions.

ii. Marks allotted are mentioned against each question/part.

iii. There is no negative marking.

iv. Do as per the instructions given.

7. **SECTION B – SUBJECTIVE TYPE QUESTIONS (26 MARKS):**

i. This section has 16 questions.

ii. A candidate has to do 10 questions.

iii. Do as per the instructions given.

iv. Marks allotted are mentioned against each question/part.

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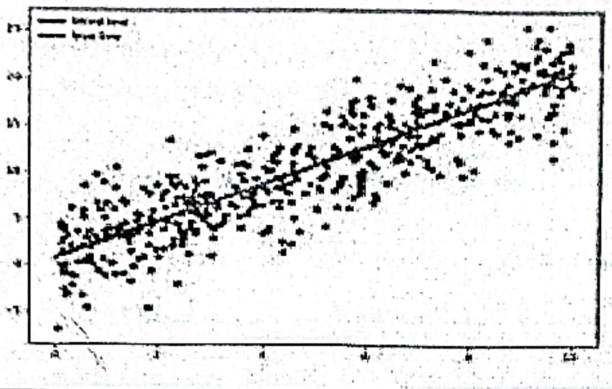
SECTION A: OBJECTIVE TYPE QUESTIONS

Q1.	Answer any 4 out of the given 6 questions on Employability Skills (1 x 4 = 4 marks)	
i.	In the communication cycle, who initiates the process? (a) Channel (b) Receiver (c) Sender (d) Time	1
ii.	Which of the following is a valid extension for Notepad file? (a) .jpg (b) .doc (c) .text (d) .txt	1

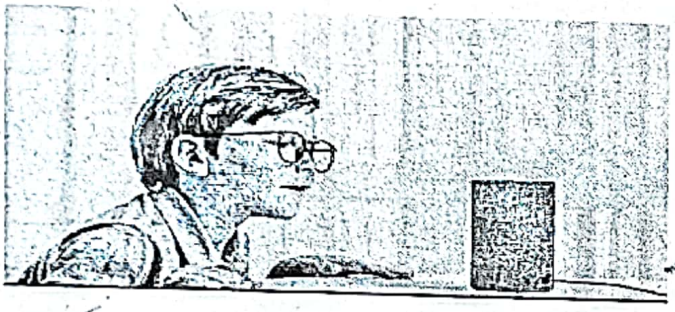
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iii.	Manisha is preparing for her exams and find herself constantly stressed due to long study hours and pressure to perform. Her teacher suggests trying yoga to manage her stress effectively. Which of the following statements is most appropriate for yoga and stress management? (a) Yoga is an effective technique for managing stress. (b) Yoga combines physical postures, breathing exercises and meditation to reduce stress levels. (c) Both (a) and (b) are correct. (d) Neither (a) nor (b) is correct.	1
iv.	Which of the following strategies is most effective for improving time management skills? (a) Working on multiple tasks simultaneously to save time. (b) Procrastinating tasks until the last minute for increased focus. (c) Prioritizing tasks based on their importance and deadlines. (d) Ignoring breaks to maximize work periods.	1
v.	Choose the option which defines sustainable development. (a) Taking care of future generations. (b) Taking care of only ourselve. (c) Taking care of ourselve and future generations. (d) Well-being of all	1
vi.	**Assertion (A):** With the growth of a business, entrepreneurs look for more people to help them. **Reason (R):** They buy more material and also hire more people to work for them. a. Both A and R are correct and R is the correct explanation of A. b. Both A and R are correct but R is not the correct explanation of A. c. A is correct but R is not correct. d. A is not correct but R is correct.	1
Q2.	**Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)**	
i.	_____ is an NLP tool to express an opinion, whether the underlying sentiment is positive, negative or neutral. (a) Text Classification (b) Machine Translation (c) Sentiment Analysis (d) Automatic Text Summarization	1
ii.	Two examples of supervised learning models are _____. (a) Classification and Regression (b) Rule Based and Learning Based (c) Clustering and Dimensionality reduction (d) Classification and Clustering	1
iii.	A retail company uses a machine learning algorithm to group similar customer purchases to improve product recommendations. Which type of algorithm is used by the company? (a) Supervised Learning (b) Unsupervised Learning (c) Reinforcement learning (d) Neural Network	1
iv.	What do framework provides in the context of problem-solving? (a) Random solutions (b) Step-by-step guidance (c) Legal advice (d) Ethical justification	1
v.	**Statement 1:** Clustering is a machine learning technique, which groups the unlabeled datasets. **Statement 2:** It is helpful for predicting numerical values based different points, such as sales revenue, projections for a given business. a. Both Statement 1 and Statement 2 are correct. b. Both Statement 1 and Statement 2 are incorrect. c. Statement 1 is correct but Statement 2 is incorrect. d. Statement 2 is correct but Statement1 is incorrect.	1
vi.	**Assertion (A):** The convolutional layer in the convolutional Neural Network (CNN) assigns importance to various features or objects in the image.	1

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	Reason (R): The main role of convolutional layer is to detect patterns such as edges, textures or shapes using filters. a. Both A and R are true and R is the correct explanation of A. b. Both A and R are true but R is not the correct explanation of A. c. A is true but R is false. d. A is false but R is true.	
Q3.	**Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)**	
i.	Observe the image showing a line best fitting the given data points. Identify the type of supervised learning model. 	1
ii.	Which of the following is the correct sequence of stages of AI project cycle? (a) Problem scoping→Data Exploration→Modelling→ Evaluation (b) Data Exploration→Problem Scoping→Modelling→Evaluation (c) Problem Scoping→Modelling→Data Exploration→Evaluation (d) Evaluation→Modelling→Data Exploration→Problem Scoping	1
iii.	An AI agent playing a game and learning from its rewards and penalties is an example of _____ learning. (a) Supervised (b) Unsupervised (c) Reinforcement (d) Evolutionary	1
iv.	In a spam email detection system, out of 1000 email received, 300 are spam. The system correctly identifies 240 spam emails as spam, but it also marks 60 legitimate emails as spam. What is the precision of the system? (a) 80% (b) 70% (c) 75% (d) 90%	1
v.	Which of the following tasks is commonly performed using Natural Language Processing? (a) Image Classification (b) Speech Recognition (c) Text Summarization (d) Object Detection	1
vi.	Which of the followingis a primary task of computer vision in artificial intelligence? (a) Speech Recognition (b) Image Classification (c) Machine Translation (d) Text-to-speech synthesis	1
Q4.	**Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)**	
i.	How is the relationship between model performance and accuracy described? (a) Inversely proportional (b) Not related (c) Directly proportional (d) Randomly fluctuating	1
ii.	Artificial Intelligence (AI) is a branch of computer science that aims to build system capable of performing tasks that typically require human intelligence. Which of the following is not a characteristic of artificial intelligence? (a) Ability to learn from experience (b) Ability to perform task without human intervention (c) Ability to feel emotions like humans (d) Ability to recognize patterns	1

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iii.	Which of the following statement is NOT true about supervised learning? (a) Require labeled data for training. (b) Used for classification and regression tasks. (c) Can be less efficient for large datasets. (d) Often used in image recognition applications.	1
iv.	Which of the following is NOT a commonly used metric to evaluate the performance of a classification model? (a) Accuracy (b) Precision (c) Mean Squared Error (d) Recall	1
v.	**Statement 1:** Unsupervised learning helps in identifying patterns and relationships with in unlabeled data. **Statement 2:** Unsupervised learning is used to train an AI agent using rewards and penalties. a. Both Statement 1 and Statement 2 are correct. b. Both Statement 1 and Statement 2 are incorrect. c. Statement 1 is correct but Statement 2 is incorrect. d. Statement 2 is correct but Statement1 is incorrect.	1
vi.	Observe the image and identify which of the following represents an example of Artificial Intelligence in daily life.  (a) Using a calculator to perform arithmetic operations (b) Writing notes with a pen (c) Voice assistant like Alexa or Siri (d) Reading a book	1
Q5.	**Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)**	
i.	In machine learning, what is the main purpose of dividing the data into training and testing sets? (a) To increase the size of the dataset. (b) To check, how well the model performs on new or unseen data. (c) To make the model run faster. (d) To remove errors from the data.	1
ii.	Which of the following is the primary goal of computer vision? (a) To enable computers to understand and interpret visual information from images or videos. (b) To improve computer hardware performance. (c) To create video games with better graphics. (d) To store large amount of data efficiently.	1
iii.	What is the process of breaking down a sentence into smaller parts like words or phrases called in Natural Language Processing(NLP)?	1
iv.	The goal when evaluating an AI model is to_____. (a) Maximize errors and minimize accuracy. (b) Minimize errors and maximize accuracy. (c) Focus solely on the numbers of data points used. (d) Prioritize the complexity of the model.	1
v.	Which of the following scenarios might have a high False Negative (FN) cost? (a) Viral disease outbreak (b) Spam	1

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	(c) Mining (d) Image search	
vi.	Which of the following is used for finding the frequency of words in some given text sample? (a) Stemming (b) Lemmatization (c) Bags of words (d) None of these	1

SECTION B: SUBJECTIVE TYPE QUESTIONS

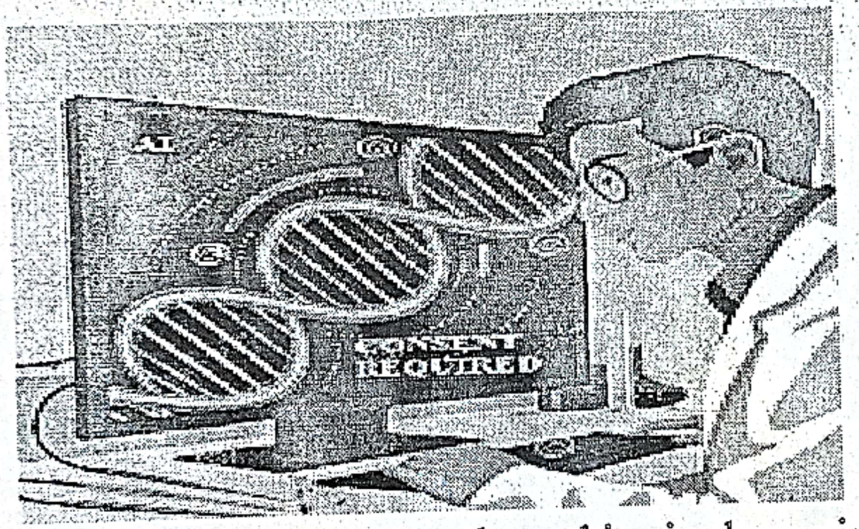
	Answer any 3 out of the given 5 questions on Employability Skills (2 x 3 = 6 marks) Answer each question in 20 – 30 words.	
Q6.	What deficit does stress management address and in what way in our lives?	2
Q7.	Rohit, a recent graduate, struggles with effective oral communication during team meetings at his new job in an Indian corporate environment. What are the three key factors that can help Rohit improve his oral communication skills in this context?	2
Q8.	What are the key benefits of agricultural entrepreneurship for rural communities?	2
Q9.	How SDG – 6 (Clean water and sanitation) and SDG – 9 (Industry Innovation and Infrastructure) are interrelated?	2
Q10.	What are temporary files and what purpose do they serve?	2
	Answer any 4 out of the given 6 questions in 20-30 words each. (2 x 4 = 8 marks)	
Q11.	What is Natural Language Processing (NLP)? Why tokenization is important in NLP?	2
Q12.	Suppose you are building a spam mail filter. Describe what False Positive and False Negative mean in this context?	2
Q13.	Describe the principle of privacy in AI and its ethical implications.	2
Q14.	Why is labeled data important for machine learning, but not always necessary for artificial intelligence?	2
Q15.	How computer vision is helping in medical imaging?	2
Q16.	Raghav uses Snapchat daily and enjoys personalized show suggestions on Netflix. He wonders how these apps understand his preference so well. Define machine learning. Also, mention any two applications of machine learning that are used in daily life.	2
	Answer any 3 out of the given 5 questions in 50– 80 words each (4 x 3 = 12 marks)	
Q17.	An AI-based system was designed to detect fraudulent bank transactions. It was tested on data set of 500 transactions, and the resulting confusion matrix is given below.	4

Confusion Matrix	Reality: Fraud Yes	Reality: Not Fraud No
Predicted: Fraud	80	40
Predicted: not fraud	30	350

(a) How many of the total cases is True Positive in the above scenario?
(b) Calculate the Precision, Recall and F1 score for fraud detection.

Q18.	Why is exploring teachable and experimental machine tools useful in learning AI?	4
Q19.	Following is the corpus: **Document 1:** "AI changes industries. It uses algorithms." **Document 2:**"AI aids healthcare. It diagnoses diseases." Perform the following functions on the above corpus. (a) Sentence segmentation (b) Tokenization (c) Stop words removal	4
Q20.	Observe the given image and answer the questions that follow.	4

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(a) What concept related to ethics is shown in this AI – medical context?

(b) How do such scenarios help students understand the importance of ethics in AI?

Q21. A machine learning model was evaluated using a confusion matrix The result were as follows: **4**

Confusion Matrix	**Predicted Positive**	**Predicted Negative**
	Yes	No
Actual Positive	50	10
Actual Negative	5	35

(a) Calculate Accuracy and Recall.

(b) What do you infer from the results?

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